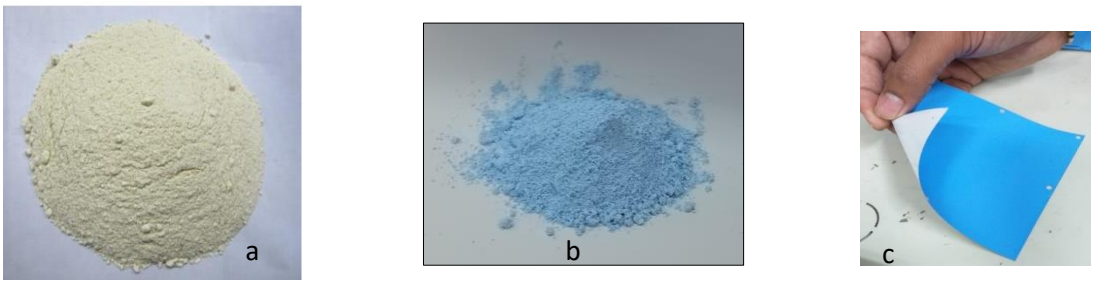
1. **Title: LTCC green tape casting**

2. **Brief Description and Product/Solution**

C-MET has developed LTCC based green tape and Ag and Ag-Pd Conductor paste for LTCC based Electronic package fabrication. LTCC has wide ranging applications, encompassing microwave circuits entering into mm wave frequencies, chip packaging, packaging of Micro Electro Mechanical Systems (MEMS), microfluidics and electronic micro systems. CMET has worked on general purpose LTCC tapes in the Cordierite system, as the glass crystallizes into Anorthite phase. The developed composition has advantages in terms of material properties especially its TCE-match to Silicon. The tape composition is a mixture of cordierite and glass in the pre-decided proportion and the LTCC mix is mixed with appropriate organic constituents to obtain the desired slurry. The slurry is then casted on a mylar film using a continuous tape caster with a doctor's blade to control the thickness of the tape. The in-house developed conductor pastes are co-firable with this LTCC tape and the shrinkage and warpage of the pastes match with those of the LTCC green tape. These tapes were also tested for their thickness uniformity and density distribution as well as for their electrical and mechanical properties. The features of the developed tapes include:

- Thickness:~100µm
- Tape casting: ~ 4 m length with 9" width
- Useful width: 7-8"
- Thickness uniformity: 1 µm across central 5" width
- Dielectric constant @100MHz: <10 (~6.0)
- Dielectric loss@100MHz: <0.15%
- Break down voltage: >1000V/25µm



Optical images of a) Cordierite powder b) LTCC mix Powder c) LTCC tapes casted at C-MET

3. **Main uses and domain**

- Bio-medical products (miniature, external testing products)
- HF communication products up to a few GHz (Mobile front-end circuits, satellite communication, remote sensing, Missiles communication)
- Internal security products (MEMS based sensor packages)
- Specialized strategic applications

4. Features and Technical Specifications

C-MET will provide one week training on the development of LTCC green tape at the premises of C-MET and casting of atleast 100 gm of slurry of will be demonstrated.

5. Platform required (if any) : Nil

6. Current Status of the System (Deployment/Product Commercialization)

Lab Level development complete. The developed LTCC was tested in-house and the properties are comparable with the application requirement.
TOT cost: 8.5 lakhs + taxes

7. Contact details for Techno-commercial Information

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